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Studierichting: Informatica
Oefeningenreeks 6D nr. 30.28

$$\sum_{n=1}^{\infty} \frac{n+1}{n(n+2)}$$

Verhoudingscriterium:

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n} &\sim \sum_{n=1}^{\infty} \frac{n+1}{n(n+2)} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{n+1}{n(n+2)}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n+1}{n(n+2)} \cdot \frac{n}{1} \\ &= \lim_{n \rightarrow \infty} \frac{n(n+1)}{n(n+2)} \\ &= \lim_{n \rightarrow \infty} \frac{n+1}{n+2} \\ &= 1 \Rightarrow \text{Divergent} \end{aligned}$$

References